Sidecar CPU overhead, Online Boutique (1,500 HTTP req/s offered to every configuration)
(a) Host CPU
Host vCPU
15.4
18.5 (+20%)
20.2 (+32%)
19.2 (+25%)
20.8 (+36%)
No sidecar
Linkerd L4
Linkerd L7
Istio L4
Istio L7
Sidecar user
Sidecar kernel
App user
App kernel
(b) Sidecar coalesces the app's traffic
App syscalls per request
936
803
661
742
614
App wakeups per request
200
183
153
172
142
Pod-to-pod packets per request
59
56
51
54
25
None
(c) Inside the sidecar
Linkerd L4
Linkerd L7
Istio L4
Istio L7
3.5
0.8
5.0
2.8
2.4
1.4
1.4
8.6
3.5
0.7
1.6
5.9
2.1
0.5
3.6
1.1
1.1
2.4
10.8
Sidecar vCPU
kernel / IPC
mTLS
L7 parsing
L7 policy
telemetry
runtime
(d) Latency, one request outstanding
Latency per HTTP request (ms)
p99
mean
18
9.0
29
14.3
41
21.0
31
15.0
54
26.8
(e) Inside the app containers
App vCPU
15.4
2.8
3.1
3.4
5.6
≈0.45
11.6
2.2
2.3
3.0
3.6
10.0
2.0
2.1
2.5
3.0
JIT / interpreted code
Runtime (GC, scheduler, VM)
RPC stack (gRPC, HTTP/2, protobuf)
Kernel
Business logic
(f) Enabling features on top of L7
Sidecar vCPU
8.6
8.6
8.8
9.2
10.5
10.8
11.0
11.1
11.9
L7
+authz
+traffic
+telem
L7 no mTLS
Linkerd
Istio
runtime
telemetry
L7 policy
L7 parsing
mTLS
kernel / IPC
jet2: Xeon 6515P, 16 cores / 32 vCPU (SMT on), 2.3 GHz fixed; 24 Online Boutique pods, same layout in every configuration;
k6 open loop from jet1, 20 s per point; latency: one k6 VU, 13 s. vCPU = cgroup CPU time of the measured window, split by perf (cycles).